7 The Singapore Mass Rapid Transit (SMRT) started its first train services in 1987. It was a massive nationwide project, beginning from the physical construction of the train tracks to the planning of the train arrival frequency. Amongst other professionals, the project involved the close collaboration of civil and structural engineers as well as transport engineers.

The Kawasaki Heavy Industries (KHI) C151 train as shown in Fig. 7.1, is Singapore's first generation of SMRT train fleet and has been in passenger service since 7 November 1987. All of the 396 KHI cars are built from 1986 to 1989 by four manufacturers in the consortium led by Kawasaki Heavy Industries.

Fig. 7.1

<u>Technical Specifications:</u>

Manufacturer:	Kawasaki Heavy Industries, Nippon Sharyo, Tokyu Car Corporation, Kinki Sharyo
Number built:	396 cars (66 trains)
Car body Construction:	Aluminium-alloy construction
Maximum Speed:	90 km h^{-1} (Design)
	80 km h^{-1} (Service)
Train Length:	138 m (for the 6 cars in one train)
Width:	3.2 m
Height:	3.7 m
Train Mass:	286000 kg (fully laden)
Doors:	1.45 m, 8 per car
Seating Capacity:	208 seats

Fig. 7.2 shows a section of an elevated MRT track with a train on it. From the structural aspect, the structure load is being supported as follows:

1. Each car, with passengers in it, has its load supported by the beam below it. Car 2 is thus supported by beam 2.
2. Car 2 and beam 2 are both supported by columns 1 and 2.

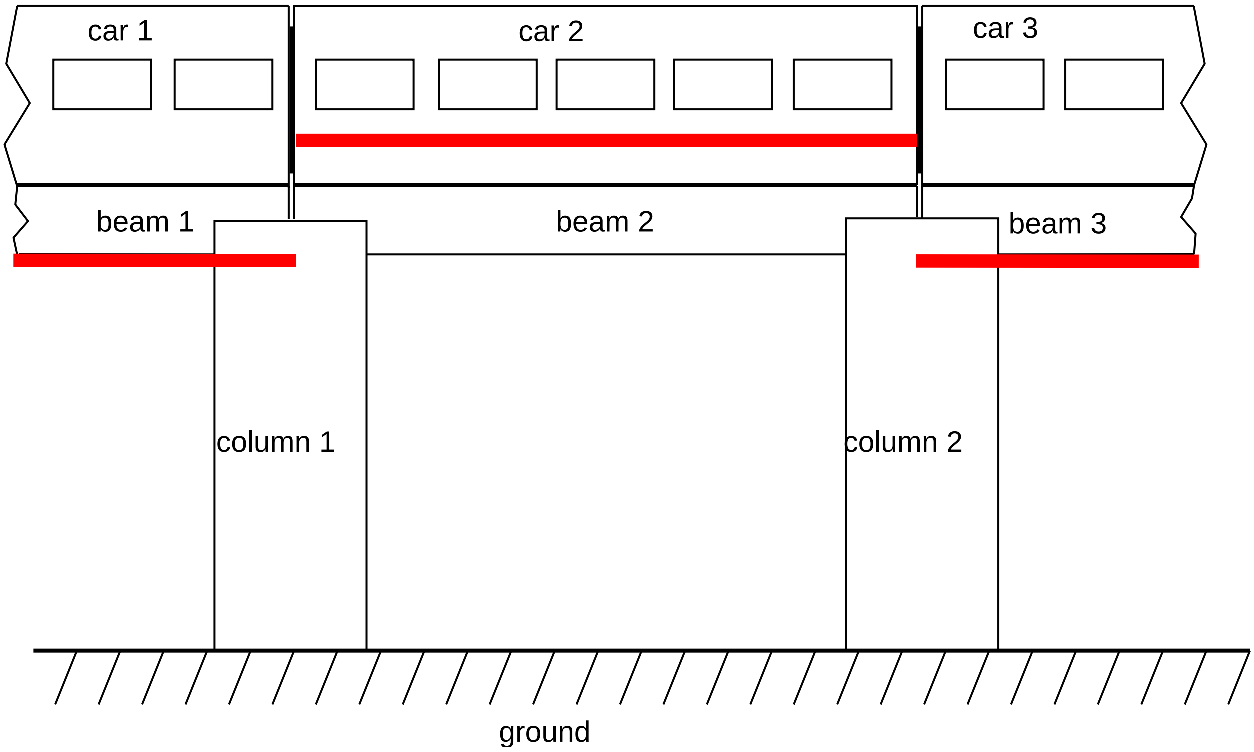

Fig. 7.2

The following set of simplified data is provided.

Weight of 1 empty car = 350 kN

Weight of 1 beam = 380 kN

Weight of 1 column = 100 kN

(a) Explain what is meant by *train arrival frequency*.

.... [1]

(b) An alloy is a combination of metals or of a metal and another element.

Suggest why trains are commonly made of aluminium alloy.

.... [1]

(c) When a train with no passengers in it, and is at the position shown in Fig. 7.2,

(i) State whether the bottom of beam 2 is under compression or tension.

.... [1]

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(ii) calculate the total normal reaction force acting on beam 2 due to the supporting columns.

normal force = ……..………........……..... N [1]

(iii) Hence, state the total load that the top of column 1 has to take.

total load = ……..………........……..... N [1]

(iv) calculate the total load that the ground directly below each column has to take.

total load = ………………........……..... N [2]

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(d) An engineer needs to design the structure such that the ground does not cave in when a fully loaded train passes overhead. In designing the structure loading, a factor of safety is incorporated.

$$\text{Factor of safety} = \frac{\text{maximum stress}}{\text{applied stress}} = \frac{\text{maximum load}}{\text{applied load}}$$

Maximum stress is defined as the maximum force the ground can withstand per unit cross-sectional area.

Applied stress is defined as the applied force the ground withstands *F*, per unit cross-sectional area *A*.

Simplified data for the applied force the ground withstands *F*, and the cross-sectional area *A*, are given in Fig. 7.3.

F / kN	A / m^2
922	4.3
916	4.4
936	4.5
958	4.6
980	4.7
996	4.8
1020	4.9
1040	5.0

Fig. 7.3

The variation with A of F is as shown in Fig. 7.4.

(i) Complete Fig. 7.4 by drawing the best-fit line (ignore any anomalous data). [1]

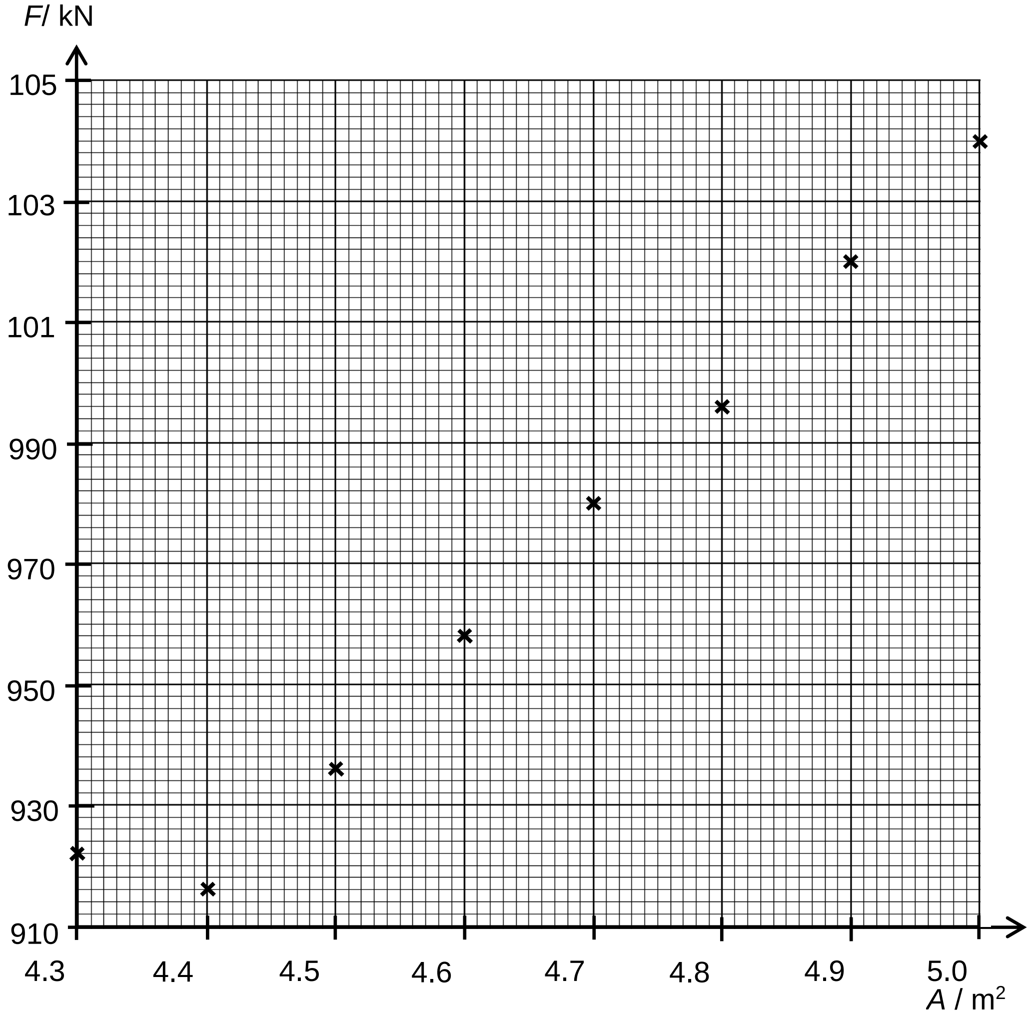

Fig. 7.4

(ii) Use Fig. 7.4 to determine the applied stress that the ground withstands.

applied stress = ……...…........……..... N m^{-2} [2]

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(iii) The column structure is considered safe if the factor of safety is greater than 2.9. Assuming that the maximum stress the ground is designed to withstand is 645 kN m^{-2}, determine whether the column structure is safe.

column structure is ......…….... [2]

(e) The simplified dimensions of each column are given in Fig. 7.5.

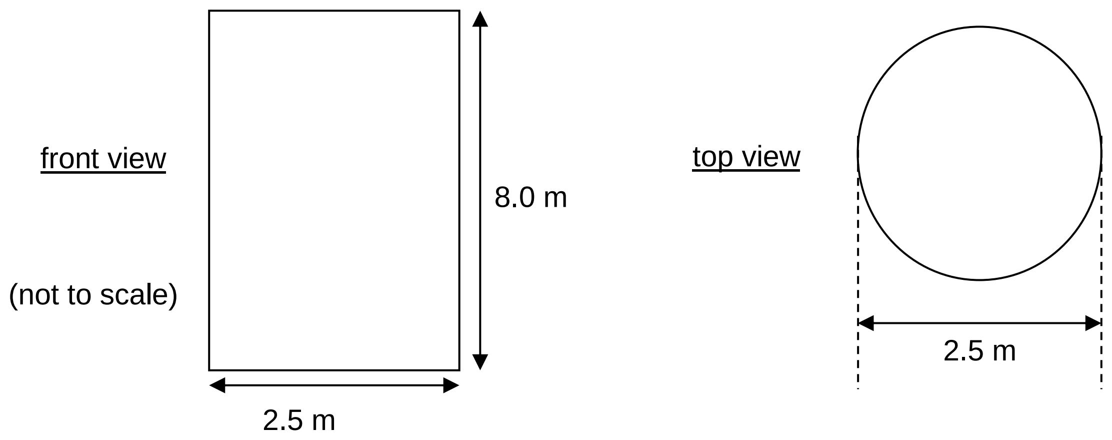

Fig. 7.5

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(i) Using the value of applied stress from **(d)(ii)**, calculate the applied load that the ground withstands.

applied load = ……………..........…….... N [2]

(ii) Hence using e(i) and c(iv), calculate the total allowable weight of passengers that each car can carry.

allowable weight = ……………..........…….... N [1]

(iii) Assuming the average mass of 1 passenger to be 60 kg and value of g to be 10 m s^{-2}, calculate the allowable number of passengers that a car can carry at any one time.

number of passengers = …………..…........…….... [2]

(f) A transport engineer is employed to design the frequency of the trains arriving at Tuas Crescent MRT Station. In order not to cause the ground to sink, he needs to look into the allowable passengers that each car can take and not overload each car. The following information is available to him:

<u>Peak hours at Tuas Crescent MRT Station</u>

Average number of east-bound passengers per minute = 240

On average, a typical east-bound train of 6 cars is anticipated to be already 75% filled just before it arrives at Tuas Crescent MRT Station.

Assuming that each car takes equal number of passengers and all board the train, determine the maximum time interval between arrival of consecutive east-bound trains at the station during peak hours.

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Maximum time interval = …….…… minutes [3]

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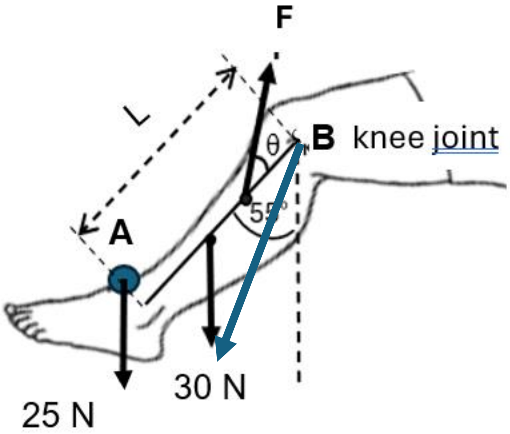
Answers and Marking Scheme

1(a)	1. The resultant force is zero. 2. The resultant moment about any axis is zero.	[B1] [B1]
(bi)	(25 x 0.40 sin 55) + (30 x 0.20 sin 55) = (F sin 25) x 0.10 F = 310 N	[M1] [A1]
(bii)	 Correctly drawn (direction) and labelled	[B1]
(biii)	Work done = increase in GPE one leg Work done =25 (L – L cos 55) + 30(L/2 – L/2 cos 55) 25 (0.4)(1- cos 55) + 30(0.2)(1- cos 55) = 6.8 J	[M1] [A1]
(biv)	Work done = (6.8)/5 = 1.4 W	[B1]

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2 **(a)** **(i)** Since the direction of the force exerted by the external agent is opposite to the direction of displacement of the mass [B1] when it moves from infinity to a point in the gravitational field [B1], the work done by the external agent is negative. Hence gravitational potential at that point is negative.

(iii)

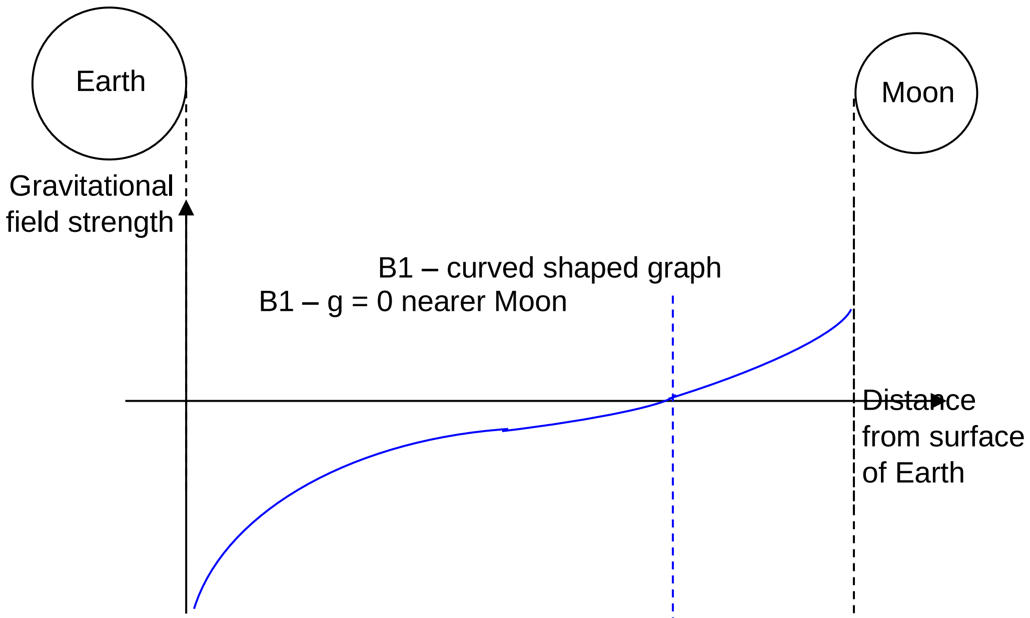

Fig. 2.1

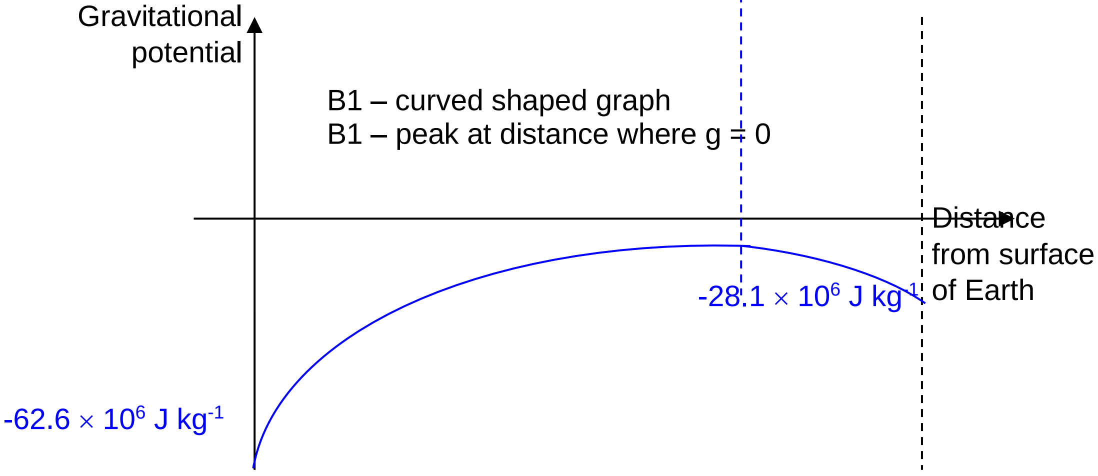

Fig. 2.2

(b)

(iv)

$$g = \frac{GM}{r^2} = \frac{(6.67\times10^{-11})(6.4\times10^{23})}{(3.4\times10^{6})^2}[M1] = 3.69\ \text{N kg}^{-1} = 3.7\ \text{N kg}^{-1}[A0]$$

(v)

Since the height of 1800 m is much smaller than the radius of the planet, it can be assumed that the stone is moved through 1800 m in a uniform gravitational field. [M1]

Hence $\Delta E_p = mg\Delta h = 2.4(3.7)(1800) = 1.6\times10^4$ J [A1]

(vi)

Assumption: All the loss in gravitational potential energy of the rock is transferred to its kinetic energy.
Total final KE + GPE = Total initial KE + GPE [M1]

$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{8r}\right) = 0 + 0 [M1]$$

[

$$\Rightarrow v = \sqrt{\frac{GM}{4r}} = \sqrt{\frac{(6.67\times10^{-11})(6.4\times10^{23})}{4(3.4\times10^{6})}} = 1.77\times10^3 \text{ m s}^{-1} [A1]$$

3	**(a)**	**Waves** travel along string and gets **reflected at the fixed point** (A/B/end) **Incident and reflected waves**, having the **same frequency, amplitude** and **speed** moving in opposite direction **superpose/interfere**	B1 B1
	(b)	Line of approximate sinusoidal shape with maximum downward displacement at P and zero displacement at each node (dots)	B1
	(c)	$\lambda = v\times T = 35\times0.04 = 1.4$ m $\Rightarrow$ distance AB $= 2.5\lambda = 2.5\times1.4 = 3.5$ m	C1 A1
	(d)(i)	No of cycles or periods $= \dfrac{0.060}{0.040} = 1.5$ Amplitude of oscillation = $\dfrac{72}{6} = 12$ mm	C1 A1
	(d)(ii)	0.0 s, 0.020 s, 0.040 s, 0.060 s, …….	A1
	(d)(iii)	$v_{max} = \omega x_o = \dfrac{2\pi}{T}x_o = \dfrac{2\pi}{0.040}\times0.012$ $= 1.89$ m s^{-1}	M1 A1
	(e)	Point P and Q are anti-phase (or 180° out of phase) Amplitude of Q < amplitude of P but their frequency is equal	B1 B1

4	**(a)**	very high/infinite resistance at low voltages (0 to 1.8V) resistance decreases as V increases (to about 90 Ω at 2.7 V)	B1 B1
	(b)(i)	From graph, current = 0.0100 A V = IR = 0.0100 (50) = 0.50 V	M1
	(b)(ii)	$V_{//} = 2.5 + 0.50 = 3.0$ V current through 75 Ω $= \dfrac{3.0}{75} = 0.040$ A current through 100 Ω $= 0.010 + 0.040 = 0.050$ A Emf $= 3.0 + (0.050)(100) = 8.0$ V	C1 C1 A1

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	(iii)	$E = Nhf$ $Pt = \dfrac{Nhc}{\lambda}$ $\dfrac{N}{t} = \dfrac{P\lambda}{hc} = \dfrac{(2.5 \times 0.0100)(4.7 \times 10^{-7})}{6.63 \times 10^{-34}(3.00 \times 10^{8})}$ $= 5.91 \times 10^{16}\ \text{s}^{-1}$ $= 5.9 \times 10^{16}\ \text{s}^{-1}$	C1 A1
	(c)	In darkness, LDR's resistance is very large, thus p.d. across parallel branch is large and this turns the LED on. When the LED is turned on, resistance of LDR decreases and p.d. across parallel branch decreases. This turns the LED off. (this cycle of LED switching on and off is repeated)	B1 B1

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5	(a) (i)	$s_x = u_x t$ $0.12 = (6.0 \times 10^{7})t$ $t = 2.0 \times 10^{-9}$ s	M1 A0
	(a) (ii)	$s_y = u_y t + \frac{1}{2}a_y t^2$ $x = 0 + \frac{1}{2}\left(\frac{qE}{m}\right)t^2$ $x = 0 + \frac{1}{2}\left(\frac{qV}{md}\right)t^2$ $= \frac{1}{2}\left(\frac{1.60 \times 10^{-19}(4000)}{9.11 \times 10^{-31}(0.08)}\right)(2.0 \times 10^{-9})^2$ $= 0.0176 = 0.018$ m	 C1 C1 A1
	(b) (i)	Component of velocity <u>perpendicular to magnetic field</u> experiences a magnetic force which provides for centripetal force for circular motion. Component of velocity <u>parallel to magnetic field</u> experiences no force, thus electron continues to move in this direction at constant speed. (combined motion is that of a helix)	B1 B1
	b(ii)	$T = \dfrac{2\pi r}{v_{\perp}}$ $= \dfrac{2\pi \times 0.029}{5.0 \times 10^{5} \sin 25°}$ $= 8.62 \times 10^{-7} = 8.6 \times 10^{-7}$ s (2sf)	M1 A0
	(b) (iii)	$d = v_{//}T$ $= (5.0 \times 10^{5} \cos 25°)(8.62 \times 10^{-7})$ $= 0.391\ \text{m} = 0.39$ m (2sf)	 C1 A1

6 **(a)(i)** 50×10^{-3} T — B1

(a)(ii) Flux linkage = $NBA = 150 \times 50 \times 10^{-3} \times 0.40 \times 10^{-4}$ — C1

$= 3.0 \times 10^{-4}$ Wb — A1

(b)(i) Final flux linkage = $150 \times 8.0 \times 10^{-3} \times 0.40 \times 10^{-4} = 4.8 \times 10^{-5}$ Wb

Change in flux linkage = final – initial = $4.8 \times 10^{-5} - 3.0 \times 10^{-4}$

$= -2.52 \times 10^{-4}$ Wb — A1

(b)(ii) Average induced e.m.f. = $\dfrac{\Delta\phi}{\Delta t} = \dfrac{2.52 \times 10^{-4}}{0.30}$ — M1

$= 8.4 \times 10^{-4}$ — A1

(c) Flux linkage decreases as the distance x increases. — C1

Hence, speed of coil need to increase to keep rate of change of flux (induced e.m.f.) constant — A1

(d) Induced current in the coil produces a magnetic field in the coil./ Induced current is in the field of the magnet. — B1

This field/The current interacts with the magnetic field of the magnet to produce a force , — B1

which opposes the motion of the coil. — B1

Hence, work need to be done to move the magnet along the axis

7 **(a)** It is the number of trains arriving at a station per unit time. — A1

(b) Aluminium alloy has high strength-to-weight ratio, thus reduces the amount of friction by reducing the weight of the trains. It has high corrosion resistance. Aluminium's natural passivation process in which a thin aluminium oxide layer forms when the metal is exposed to oxygen, reduces the possibility of further oxidation. — A1

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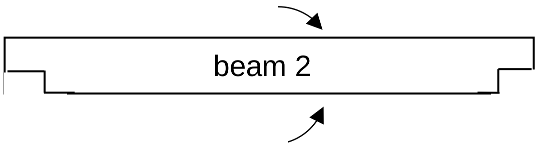
(c) **(i)** top of beam — A1

top of beam under compression



bottom of beam under tension

(ii) Total normal reaction forces = $(350 + 380) \times 10^{3}$

$= 7.30 \times 10^{5}$ N — A1

(iii) Total load column 1 has to take = 7.30×10^{5} N — A1

(iv) Total load ground has to take = $(730 + 100) \times 10^{3}$ — M1

$= 8.30 \times 10^{5}$ N — A1

(d) **(i)** Coordinate (4.3, 922) is treated as anomaly. Best fit line drawn through the rest of the seven points. — A1

(ii) Gradient of line $= \dfrac{1040 - 916}{5.00 - 4.40}$ M1

$= 2.07 \times 10^5$ N m^{-2} A1

(iii) Factor of safety = $\dfrac{645 \times 10^3}{207 \times 10^3}$

$= 3.12$ M1

Since factor of safety is greater than 2.9, it is safe. A1

(e) **(i)** Applied load = $(207 \times 10^3)\pi(\dfrac{2.5}{2})^2$ M1

$= 1.0 \times 10^6$ N A1

(ii) Total allowable weight of passengers = $(1016 – 830) \times 10^3$

$= 1.9 \times 10^5$ N A1

(iii) Total allowable number of passengers per car = $(\dfrac{1.9 \times 10^5}{60 \times 10})$ M1

$= 316$ A1

(f) Number of passengers a car can take when train arrives at station

$= 0.25 \times 316 = 77.5$ (25% of the total allowable) C1

Total number of passengers train can take = 79×6

$= 474$ M1

Longest time interval between train arrivals = $(\dfrac{474}{240})$

$= 1.98$ minutes A1

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